

Ex. No.: 11c)

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### OPTIMAL

Aim:

To write a c program to implement Optimal page replacement algorithm.

ALGORITHM:

- 1.Start the process
- 2.Declare the size
- 3.Get the number of pages to be inserted
- 4.Get the value
- 5.Declare counter and stack
- 6.Select the least frequently used page by counter value
- 7.Stack them according the selection.
- 8.Display the values
- 9.Stop the process

PROGRAM:

```
#include <stdio.h> int predict(int pages[], int frames[], int n, int
index, int frameSize) {      int res = -1, farthest = index; for (int
i = 0; i < frameSize; i++) {      int j;      for (j = index;
j < n; j++) {      if (frames[i] == pages[j]) {      if (j >
farthest) {      farthest = j;      res = i;
      }
      break;
      }
      }
      if (j == n)
      return i;
      }
      return (res == -1) ? 0 : res;
} int main()
{
      int frames[10], pages[30];      int
i, j, k, n, frameSize, faults = 0;      int
hit;
```

```

        printf("Enter number of frames: ");
scanf("%d", &frameSize);    printf("Enter
number of pages: "); scanf("%d", &n);
        printf("Enter reference string:
");    for (i = 0; i < n; i++)
scanf("%d", &pages[i]);    for (i = 0;
i < frameSize; i++)    frames[i]
= -1;    printf("\n");    for (i = 0; i < n;
i++) {        hit = 0;        for (j = 0;
j < frameSize; j++) {            if
(frames[j] == pages[i]) {                hit
= 1;                break;
            }
        }
        if (!hit) {            int
empty = -1;            for (j = 0; j <
frameSize; j++) {                if
(frames[j] == -1) {
empty = j;                break;
            }
        }        if (empty != -1) {
frames[empty] = pages[i];
        } else {            int pos = predict(pages, frames, n, i
+ 1, frameSize);            frames[pos] = pages[i];
        }
faults++;
    }
    for (k = 0; k < frameSize; k++)
{        if (frames[k] != -1)
printf("%d ", frames[k]);
else        printf("-1 ");
    }
    printf("\n");
}

printf("\nTotal Page Faults = %d\n", faults);
return 0;
}

```

## OUTPUT:

```
Enter number of frames: 3
Enter number of pages: 10
Enter reference string: 3
2 3 1 4 2 5 3 2 8 6
3 -1 -1
3 2 -1
3 2 6
3 2 8
3 2 8
4 2 8
1 2 8
1 2 8
1 2 8
6 2 8
Total Page Faults = 7
```

## RESULT:

Hence, page faults that occur using OPTIMAL page replacement technique has been found.